

Roadmap - Topic 3 Human Body Systems

Learning Objectives	Vocabulary			Questions
Lesson 1: Body Organization				
- list the levels of organization in the body - describe the organization of body systems - describe the function of cells, tissue, organs, and body systems - compare the structure and function of a body system to other systems - explain the general functions of body systems, including how they work together to function	Organ system Tissue Organ Skeletal Integumentary	Muscular Respiratory Digestive Excretory Nervous	Endocrine Reproductive Immune	1. How do the organs of an organ system contribute to the organ system’s function? 2. Why does the human body need multiple organ systems to function? 3. Which systems are used for transport materials within the body?
Lesson 2: Systems Interacting				
- describe and explain stimulus-response situations - how the glands of the endocrine system control body processes - how the nervous system compares to the endocrine system - how systems interact to maintain homeostasis	Stimulus Response Gland Hormone Stress	Homeostasis		1. What are some of the internal conditions maintained by homeostasis? 2. What is an example of a cycle that the body performs to maintain homeostasis?
Lesson 3: Supplying Energy				
- explain the important nutrients a body needs to carry out its processes - how the body’s systems process the foods you eat - why some food choices are healthier than others - how food becomes materials the body can use	Digestion Nutrients Carbohydrates Peristalsis Saliva	Enzyme Absorption Protein Fats Vitamins	Minerals Mechanical digestion Chemical digestion Villi	1. What organs are involved in mechanical digestion? Chemical digestion? 2. Why does digestion happen before absorption? 3. Why does the small intestine need to have a huge surface area?

Lesson 4: Managing Materials				
- how body systems interact to transport materials throughout the body - how the respiratory system interacts with other systems to exchange gases - how other body systems interact with the excretory system to remove wastes	Circulatory Artery Capillary Vein	Lymph Bronchi Alveoli Excretion Nephron	Cardiovascular Heart Blood pressure Diaphragm Cellular respiration	1. How do the respiratory and circulatory systems work together to provide oxygen to the body? 2. How does the lymphatic system support the cardiovascular system? 3. What are the functions of the different organs in the excretory system?
Lesson 5: Controlling Processes				
- what systems control processes in the human body - how nerve signals travel - how the body senses and reacts to surroundings	Neuron Synapse Brain Spinal Cord Gland	Negative feedback Reflex Impulse Central	Peripheral Somatic Autonomic Hypothalamus Pituitary	1. How does an impulse pass from one neuron to the next? 2. What functions are controlled by the somatic nervous system? The autonomic nervous system?